

Sensitivity lists

An always block does not run on every tick of simulation time

What wakes an always block

- For combinational logic
- Classic Verilog wrote at open-paren A or B close-paren
- If you list only A but also read B
- For sequential logic, posedge clk means the block runs only on a rising clock edge

Browser lab

igital Design and Verification Platform

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Quiz: sensitivity: The sensitivity list of an always block names...

☐ which signal events re-run the block

☐ only the outputs the block drives

☐ the module's port list

☐ synthesis timing constraints

Show hint

Check

Next

Load challenge setup

Idle

Quiz: sensitivity

Quiz: @(*)

Quiz: incomplete

Quiz: posedge

Quiz: async reset

Quiz: @(clk)

Quiz: always_comb

Quiz: sim vs synth

Pick: AND incomplete

Pick: AND @(*)

Pick: mux incomplete

Pick: true FF

Pick: async list

See: stale B

See: @(*) tracks B

See: mux data miss

See: D ignored

See: sample on rise

See: async clear

See: forgotten async

See: mux tracks D0

Quiz: or vs ,

Sensitivity explorer

Load starter exampleClear log

FUNCTION

Combo AND

Intent: Y = A & B (combinational)

@(A or B)

@(A) incomplete

@(*)

always_comb

Complete combo sensitivity

A

B

```
always @(A or B) begin
  Y = A & B;
end
```

Both RHS reads are in the list — any change of A or B recomputes Y.

Poke signals

A=1

B=1

Y=1

Event log

Starter: Y = A & B with always @(A or B). Toggle A or B — both make the block.

Sensitivity list lab starter

Real Verilog practice

```
wsl - module06-sensitivity-list/examples/navigation

# Track A - real Unix

# real Verilog session (Track A)

$ cat examples/sensitivity/sensitivity.v
// sensitivity.v - combo list vs clock edge
module sens_demo(
    input    A,
    input    B,
    input    clk,
    input    D,
    output reg Y,
    output reg Q
);
    always @(A or B)
        Y = A & B;

    always @(posedge clk)
        Q ≤ D;
endmodule

$ iverilog -t null examples/sensitivity/sensitivity.v
(syntax check passed)
```

Real Verilog — combo and clocked sensitivity

Pitfalls to watch

- Do not hand-write or lists and forget a signal you read
- Do not use level-sensitive at open-paren clk close-paren when you mean a flip-flop
- At-star fixes combo lists in modern Verilog
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter, try the incomplete list, and poke the flip-flop scenario
- On real Verilog
- When you are ready, take the short quiz, then continue to latch risk in the next module